

PAPER_679: Aether Superfluid Dynamics: UQFF Universal Aether as a Bosonic Condensate

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Framework: Unified Quantum Field Framework (UQFF) v5.30

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Domain: Quantum Field Theory / Superfluids

Abstract

The UQFF Universal Aether [UA] is modelled as a coherent bosonic superfluid described by a macroscopic wavefunction $\Psi(r, t) = \sqrt{n_{\text{UA}}}e^{i\phi}$. We derive the healing length $\xi_{\text{UA}} = \hbar/\sqrt{2m_{\text{UA}}g_{\text{UA}}n_{\text{UA}}}$, sound speed $c_{\text{UA}} = \sqrt{g_{\text{UA}}n_{\text{UA}}/m_{\text{UA}}}$, and effective gravitational coupling $g_{\text{eff}}(r) = g_N(r)(1 + c_{\text{UA}}^2/c^2 \cdot f_{\text{TRZ}}\rho_{\text{UA}}/\rho_{\text{SCm}})$.

Primary UQFF Equation

$$c_{\text{UA}} = \sqrt{\frac{g_{\text{UA}}n_{\text{UA}}}{m_{\text{UA}}}}, \quad \xi_{\text{UA}} = \frac{\hbar}{\sqrt{2m_{\text{UA}}g_{\text{UA}}n_{\text{UA}}}}$$

Parameters

Parameter	Value	Description
m_{UA}	$\sim 2 \times 10^{-68} \text{ kg}$	Ultralight boson mass
ρ_{UA}	$7.09 \times 10^{-36} \text{ kg/m}^3$	Aether density
g_{UA}	$10^{-10} \text{ m}^3/\text{s}^2$	Interaction strength

UQFF Constants

Constant	Value	Description
ρ_{UA}	$7.09 \times 10^{-36} \text{ kg/m}^3$	Universal Aether density
ρ_{SCm}	$7.09 \times 10^{-37} \text{ kg/m}^3$	Schwarzschild condensate
f_{TRZ}	0.1	Time-reversal zone factor
κ	$5 \times 10^{-4} \text{ day}^{-1}$	UQFF calibration constant
[SSq]	0.57	Superstring quenching factor
μ_J	$3.38 \times 10^{23} \text{ J}\cdot\text{m}$	Magnetic string coupling
γ	$5 \times 10^{-5} / 86400 \text{ s}^{-1}$	Aether oscillation decay

C++ Implementation

```
#include "AetherSuperfluidDynamics.h"

AetherSuperfluidDynamics obj;
auto result = obj.compute_primary(...); // primary equation
obj.simulate_over_M(M_start, M_end, dM, "output.csv");
```

Python CP4 Calculator

```
from CondensedPhysics2 import AetherSuperfluidDynamicsCalculator

calc = AetherSuperfluidDynamicsCalculator()
result = calc.compute() # returns all UQFF quantities
print(result)
```

References

Penrose 1996 (quantum collapse); Sudarsky 2020 (BEC dark matter); UQFF Session 173

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